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1. A box contains 6 red balls and 4 green balls. Two balls are drawn at random, one after the other, without replacement. what is the probability that the second ball drawn is green, given that the first ball drawn as red?

Given that Red balls = 6

Green balls = 4

Total balls = 6+4 = 10

∵ the first ball drawn is red

remaining red balls = 6-1 = 5

remaining green balls = 4 (unchanged)

Total remaining balls = 10-1 = 9

Now, the probability that the second ball drawn is green

is $P\left(\frac{\text{Second is green}}{\text{First is Red}}\right) = \frac{4}{9}$.

So, The conditional probability that the second ball drawn is green, given that the first ball drawn was red is $\frac{4}{9}$

2. A factory produces 3 types of machines : Type A, Type B, and Type C. The probability of producing each type of machine is as follows $P(A) = 0.5$, $P(B) = 0.3$, $P(C) = 0.2$. The probability of a machine being defective given its type is $P(D|A) = 0.02$, $P(D|B) = 0.05$ and $P(D|C) = 0.01$. A machine is selected at random, and it is found to be defective. what is the probability that the defective machine is of Type A

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Given $P(A) = 0.5$, $P(B) = 0.3$, $P(C) = 0.2$

probability of selecting a defective item machine

machine A is $P(D/A) = 0.02$

machine B is $P(D/B) = 0.05$

machine C is $P(D/C) = 0.01$

Now By Baye's theorem

the probability that the defective machine is of Type A

$$P(A/D) = \frac{P(A) \cdot P(D/A)}{P(A) \cdot P(D/A) + P(B) \cdot P(D/B) + P(C) \cdot P(D/C)}$$

$$= \frac{0.5 \times 0.02}{0.5 \times 0.02 + 0.3 \times 0.05 + 0.2 \times 0.01}$$

$$= \frac{0.01}{0.01 + 0.015 + 0.002}$$

$$= \frac{0.01}{0.027}$$

$$= 10/27 = 0.3703\ldots$$

3. A bag A contains 2 white and 3 red balls and bag B contains 4 white and 5 red balls. one ball is drawn at random from one bags and it is found to be red. Find the probability that the red ball drawn is from bag B.

Let A be the event of choosing Bag A, and B be the event of choosing Bag B.

Since one bag is chosen at random:

$$P(A) = \frac{1}{2} \text{ and } P(B) = \frac{1}{2}$$

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Let R be the event of drawing a red ball.

Bag A contains 2 white and 3 red balls (Total = 5)

$$P(R/A) = \frac{3}{5}$$

Bag B contains 4 white and 5 red balls (Total = 9)

$$P(R/B) = 5/9$$

By Baye's Theorem

$$P(B/R) = \frac{P(R/B) \cdot P(B)}{P(R/A) \cdot P(A) + P(R/B)\ P(B)}$$

$$= \frac{\frac{5}{9} \cdot \frac{1}{2}}{\frac{3}{5} \cdot \frac{1}{2} + \frac{5}{9} \cdot \frac{1}{2}} = \frac{\frac{5}{9}}{\frac{3}{5} + \frac{5}{9}} = \frac{25}{52}$$

∴ The probability that the red ball drawn is from bag B is $\frac{25}{52}$

4. A business man goes to hotels X, Y, Z, 20%., 50% and 30% of the time respectively. It is shown that 5%, 4% and 8% of the rooms in X, Y, Z hotels have faulty plumbing. What is the probability that business man's room having faulty plumbing is assigned to hotel X?

The probability that businessman goes to hotel X, $P(X) = \frac{20}{100} = 0.2$

The probability that businessman goes to hotel Y, $P(Y) = \frac{50}{100} = 0.5$

The probability that businessman goes to hotel Z, $P(Z) = \frac{30}{100} = 0.3$

If A is the event of faulty plumbing

The probability that faulty plumbing is in hotel X, $P(A/X) = \frac{5}{100} = 0.05$

The probability that faulty plumbing is in hotel Y, $P(A/Y) = \frac{4}{100} = 0.04$

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The probability that the faulty plumbing is in hotel z, $P(A/z) = \frac{8}{100} = 0.08$

Now,

The probability that bussinessman's room having faulty plumbing is assigned to hotel x is

$$P(x/A) = \frac{P(x) \cdot P(A/x)}{P(x) \cdot P(A/x) + P(y) \cdot P(A/y) + P(z) \cdot P(A/z)}$$

$$= \frac{0.2 \times 0.05}{(0.2 \times 0.05) + (0.5 \times 0.04) + (0.3 \times 0.08)}$$

$$= \frac{0.01}{0.01 + 0.02 + 0.024}$$

$$= \frac{0.01}{0.054} = 0.185$$

5. Suppose 5 men out of 100 and 25 women out of 10,000 are colourblind. A colour blind person is chosen at random. what is the probability that the person being male?. (Assume male and female to be in equal members.)

Let M be the event that the chosen person is Male.

Let F be the event that the chosen person is female

Let C be the event that the chosen person is colourblind

Given male and female populations are equal

So, $P(M) = P(F) = 0.5$

probability that a man is colourblind : $P(C/M) = \frac{5}{100} = 0.05$

probability that a woman is colourblind

$$P(C/F) = \frac{25}{10,000} = 0.0025$$

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Now, the probability that the person is male

$$P(M/C) = \frac{P(M) \times P(C/M)}{P(M) \times P(C/M) + P(F) \times P(C/F)}$$

$$= \frac{0.5 \times 0.05}{(0.5 \times 0.05) + (0.5 \times 0.0025)}$$

$$= \frac{0.025}{0.02625}$$

$$= \frac{2500}{2625} = \frac{20}{21} \approx 0.95238$$

6. A random variable X has the following probability function

X	0	1	2	3	4	5	6	7
$P(x)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2+K$

determine (i) K (ii) $P(x<6)$, $P(x \geq 6)$ and $P(0<x<5)$

(iii) If $P(x \leq k) > 1/2$, find the minimum value of k

(iv) Standard Deviation of X and v) cummulative distribution of X

We know that

total probability distribution = 1

So $\sum P(x) = 1$

$$0 + K + 2K + 2K + 3K + K^2 + 2K^2 + 7K^2 + K = 1$$

$$10K^2 + 9K = 1$$

$$10K^2 + 9K - 1 = 0$$

$$10K(K+1) - 1(K+1) = 0$$

$$(10K-1)(K+1) = 0$$

$$K = \frac{1}{10} \text{ or } 0.1 \text{ or } K = -1$$

Since probability cannot be negetive $K = 0.1$

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(ii) $P(x < 6)$

$P(x < 6) = P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4) + P(x=5)$

$= 0 + K + 2K + 2K + 3K + K^2$

$= 8K + K^2$

$= 8(0.1) + (0.1)^2$

$= 0.81$

(iii) $P(x \geq 6)$

$P(x \geq 6) = P(x=6) + P(x=7)$

$= 2K^2 + 7K^2 + K$

$= 9K^2 + K$

$= 9(0.1)^2 + 0.1$

$= 0.19$

$P(0 < x < 5)$

$P(0 < x < 5) = P(x=1) + P(x=2) + P(x=3) + P(x=4)$

$= K + 2K + 2K + 3K$

$= 8K$

$= 8(0.1)$

$= 0.8$

(iii) $P(x \leq K) > 1/2$

$P(x \leq 0) = 0$

$P(x \leq 1) = 0.1$

$P(x \leq 2) = 0.3$

$P(x \leq 3) = 0.5$

$P(x \leq 4) = 0.8$

so at $x = 4$ $P(x \leq K) > 1/2$

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(iv) standard deviation

We know

variance $\sigma^2 = E(x^2) - [E(x)]^2$

$E(x^2) = \sum x^2 \cdot P(x)$

$= (0^2 \times 0) + (1^2 \times 0.1) + (2^2 \times 0.2) + (3^2 \times 0.2) + (4^2 \times 0.3)$
$+ 5^2(0.01) + (6^2 \times 0.02) + 7^2(0.17)$

$= 16.8$

$E(x) = \sum x \cdot P(x)$

$= 0 \times 0 + 1 \times 0.1 + 2 \times 0.2 + 3 \times 0.2 + 4 \times 0.3 + 5 \times 0.01 + 6 \times 0.02$
$+ 7 \times 0.12$

$= 3.66$

Now $\sigma^2 = 16.8 - (3.66)^2$

$= 3.4044$

So, standard deviation $\sigma = +\sqrt{\sigma^2}$

$= \sqrt{3.4044}$

$= 1.8451$

(v) Cummulative distribution function

$\left.\begin{aligned} P(0 \leq x \leq 7) &= P(x=0) + \ldots\ldots P(x=7) \\ &= 0 + 0 + 0.1 + 0.3 + 0.5 \end{aligned}\right\} x$

$P(x < 0) = 0$

$P(0 \leq x < 1) = 0.0$

$P(1 \leq x < 2) = 0.1$

$P(2 \leq x < 3) = 0.3$

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$P(3 \le x < 4) = 0.5$

$P(4 \le x < 5) = 0.8$

$P(5 \le x < 6) = 0.81$

$P(6 \le x < 7) = 0.83$

$P(x \ge 7) = 1.$

7. Two dice are thrown. Let X assigns to each point (a, b) in S the maximum of its numbers i.e. $X(a,b) = \max(a,b)$. Find the probability distribution. X is a random variable with $X(S) = \{1, 2, 3, 4, 5, 6\}$. Also find the mean and variance of the distribution.

When two dice are thrown Sample space of S as 36 outcomes

i.e., $S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)$
$(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)$
$(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)$
$(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)$
$(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)$
$(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$

X :	1	2	3	4	5	6
P(x) :	1/36	3/36	5/36	7/36	9/36	11/36

$X(a,b) = \text{Max}(a,b)$; i.e, $x = \{1, 2, 3, 4, 5, 6\}$

for $(x = 1)$ favourable cases are $(1,1)$ So, $P(x=1) = 1/36$

$(x = 2)$ favourable cases are $(1,2), (2,1), (2,2) = 3/36$

Similarly for $x = 3, 4, 5, 6$

Now Mean $\mu = \sum p_i x_i$

$$= \frac{1}{36}(1) + \frac{3}{36}(2) + \frac{5}{36}(3) + \frac{7}{36}(4) + \frac{9}{36}(5) + \frac{11}{36}(6)$$

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$= \frac{1}{36}(1+6+15+28+45+66)$

$= \frac{161}{36} \approx 4.4722$

variance $\sigma^2 = E(x^2) - [E(x)]^2$

$= \Sigma p_i x_i^2 - \mu^2$

$= \frac{1}{36}(1) + \frac{3}{36}(2^2) + \frac{5}{36}(3^2) + \frac{7}{36}(4^2) + \frac{9}{36}(5^2) + \frac{11}{36}(6^2) - \mu^2$

$= \frac{1}{36}(1+12+45+112+225+396) - \mu^2$

$= \frac{791}{36} - (4.4722)^2$

$= 21.9722 - 20.000572$

$= 1.9716$

Standard deviation $\sigma = \sqrt{\sigma^2}$

$= \sqrt{1.9716}$

$= 1.4041$

8. For the following probability distribution

X	-3	6	9
p(x)	1/6	1/2	1/3

Find (i) $E(x)$ (ii) $E(x^2)$ (iii) $E(2x+1)^2$

We know that

$E(x) = \Sigma p_i x_i$

$= \left(-3 \times \frac{1}{6}\right) + (6 \times 1/2) + (9 \times 1/3)$

$= \left(\frac{-1}{2}\right) + \left(\frac{3}{3}\right) + 3$

$= 5.5$

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(ii) $E(x^2)$

$$E(x^2) = \sum x^2 \cdot p(x)$$

$$= \left((-3)^2 \times 1/6\right) + \left(6^2 \times 1/2\right) + \left(9^2 \times \frac{1}{3}\right)$$

$$= \frac{9}{6} + \frac{36}{2} + \frac{81}{3}$$

$$= \frac{93}{2} = 46.5$$

(iii) $E(2x+1)^2$

$$(2x+1)^2 = 4x^2 + 4x + 1$$

$$E(2x+1)^2 = E(4x^2 + 4x + 1) = 4E(x^2) + 4E(x) + 1$$

$$= 4(46.5) + 4(5.5) + 1$$

$$= 186 + 22 + 1$$

$$= 209$$

9. The probability density function of a continuous random variable is given by $f(x) = \begin{cases} K(1-x^2) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

find (i) K (ii) Probability between 0.1 and 0.2 (iii) Standard deviation

we know that $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$$\Rightarrow \int_0^1 K(1-x^2)\,dx = 1$$

$$K\left[x - \frac{x^3}{3}\right]_0^1 = 1$$

$$K\left[1 - 1/3 - 0\right] = 1$$

$$K\left[\frac{2}{3}\right] = 1$$

$$\Rightarrow K = 3/2$$

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(ii) probability between 0.1 and 0.2

$$P(0.1 < x < 0.2) = \int_{0.1}^{0.2} f(x)\,dx = \int_{0.1}^{0.2} k(1-x^2)\,dx$$

$$= \frac{3}{2}\left[x - \frac{x^3}{3}\right]_{0.1}^{0.2}$$

$$= \frac{3}{2}\left[0.2 - \frac{(0.2)^3}{3} - 0.1 + \frac{(0.1)^3}{3}\right]$$

$$= \frac{0.293}{2} = 0.1465$$

(iii) Standard deviation

for calculating standard deviation we need variance

$$\sigma^2 = E(X^2) - [E(X)]^2$$

$$E(X) = \int_0^1 x \cdot f(x)\,dx = \int_0^1 x \cdot \frac{3}{2}(1-x^2)\,dx$$

$$= \frac{3}{2}\int_0^1 (x - x^3)\,dx = \frac{3}{2}\left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1$$

$$= \frac{3}{2}\left(\frac{1}{2} - \frac{1}{4}\right) = \frac{3}{8}$$

$$[E(X^2)]^2 = \int_0^1 x^2 \cdot f(x)\,dx = \int_0^1 x^2 \frac{3}{2}(1-x^2)\,dx$$

$$= \frac{3}{2}\int_0^1 (x^2 - x^4)\,dx = \frac{3}{2}\left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1$$

$$= \frac{3}{2}\left(\frac{1}{3} - \frac{1}{5}\right) = \frac{3}{2}\left(\frac{2}{15}\right) = \frac{1}{5}$$

$$\sigma^2 = \frac{1}{5} - \left(\frac{3}{8}\right)^2 = \frac{1}{5} - \frac{9}{64} = 0.059375$$

$\therefore$ Standard deviation $\sigma = \sqrt{0.059375} = 0.243689$

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10. A continuous random variable X has a pdf $f(x) = Kx^2e^{-x}$, $x \geq 0$ find K, mean and variance.

we know that In continuous random variable $\int_{-\infty}^{\infty} f(x)dx = 1$

since given $x \geq 0$ so, $\int_0^{\infty} f(x)dx = 1$

$$= \int_0^{\infty} Kx^2e^{-x}dx = 1$$

$$\int uv = u\int v - \int(du \int v dv)$$

$$K\int_0^{\infty} x^2e^{-x}dx = 1$$

u v

$$K\int_0^{\infty} x^2e^{-x}dx = \cancel{K(x^2)e^{-x}+}$$

$$= \left(\frac{x^2e^{-x}}{(-1)} - \int 2x\left(\frac{e^{-x}}{-1}\right)dx\right)K$$

$$= \left(-x^2e^{-x} + 2x\int xe^{-x}dx\right)K$$

$$= \left(-x^2e^{-x} + 2\left[\frac{xe^{-x}}{-1}\right] - \int\frac{e^{-x}}{-1}\right)K$$

$$= e^{-x}\left[x^2 + 2x + 2\right])K$$

$$K\int_0^{\infty} x^2e^{-x}dx = \left(-e^{-x}(x^2+2x+2)\right)_0^{\infty} K = 1$$

$$K \cdot 2 = 1$$

$$K = 1/2$$

(ii) Mean $\mu = E(x)$

$$E(X) = \int_0^{\infty} x \cdot f(x)dx.$$

$$= \int_0^{\infty} x\left(\frac{1}{2}x^2e^{-x}\right)dx$$

$$= \frac{1}{2}\int_0^{\infty} x^3e^{-x}dx$$

$$= \frac{6}{2} = 3$$

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Variance $\sigma^2 = E(x^2) - [E(x)]^2$

$$E(x^2) = \int_0^\infty x^2 f(x)\,dx = \int_0^\infty x^2 \cdot \tfrac{1}{2} x^2 e^{-x} = \frac{1}{2}\int_0^\infty x^4 e^{-x}\,dx$$

$$= \frac{24}{2} = 12 \quad \left[\because \text{Gamma function } \int_0^\infty x^4 e^{-x}dx = 4! = 24\right]$$

Now $\text{var}(x) = \sigma^2 = 12 - (3)^2 = 3.$

11. A continuous random variable x has the following cummulative distribution function

$$F(x) = \begin{cases} 0 & , \text{ if } x \le 1 \\ K(x-1)^3 & , \text{ if } 1 \le x \le 3, \\ 1 & , \text{ if } x > 3 \end{cases}$$

Find (i) K (ii) P.d.f of x (iii) mean

Here Given continuous random variable X with Cumulative distribution function.

So, $F(x) = K(x-1)^3$ in $1 \le x \le 3$

At upper boundary $F(3) = 1$

$$F(3) = K(3-1)^3 = 1$$

$$K(2^3) = 1$$

$$K = 1/8$$

(ii) Probability Density Function

$$f(x) = \frac{d}{dx}[F(x)] \quad \text{in} \quad (1 \le x \le 3)$$

Now $f(x) = \frac{d}{dx}\left[K(x-1)^3\right] = K3(x-1)^2 = \frac{3}{8}(x-1)^2$

Now $$f(x) = \begin{cases} 3/8\ (x-1)^2 & , \text{ if } 1 \le x \le 3 \\ 0 & , \text{ otherwise} \end{cases}$$

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Mean $\mu = E(X) = \int_{-\infty}^{\infty} x \cdot f(x)\,dx = \int_{1}^{3} x \cdot \frac{3}{8}(x-1)^2\,dx$

$= \frac{3}{8}\int_{1}^{3} (x)(x^2 - 2x + 1)\,dx$

$= \frac{3}{8}\int_{1}^{3} (x^3 - 2x^2 + x)\,dx$

$= \frac{3}{8}\left[\frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2}\right]_1^3$

$= \frac{3}{8}\left[\frac{3^4}{4} - \frac{2(3^3)}{3} + \frac{3^2}{2} - \frac{1}{4} + \frac{2}{3} - \frac{1}{2}\right]$

$= \frac{3}{8}\left[\frac{81}{4} - \frac{54}{3} + \frac{9}{2} - \frac{1}{4} + \frac{2}{3} - \frac{1}{2}\right]$

$= 2.5$

12. Let the random variable X have the p.d.f $f(x) = \begin{cases} \frac{1}{2}xe^{-x/2}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$

find the moment generating function, mean & variance of X

Sol Moment generating function

$M_X(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x)\,dx$

$M_X(t) = \int_{0}^{\infty} e^{tx}\left(\frac{1}{2}e^{-x/2}\right)dx$

$= \frac{1}{2}\int_{0}^{\infty} e^{-\left(\frac{1}{2}-t\right)x}\,dx$

$= \frac{1}{2}\left[\frac{e^{-\left(\frac{1}{2}-t\right)x}}{-\left(\frac{1}{2}-t\right)}\right]_0^{\infty}$

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$$M_X(t) = \frac{1}{2}\left[0 - \frac{1}{-\left(\frac{1}{2}-t\right)}\right]$$

$$M_X(t) = \frac{1/2}{1/2 - t} = \frac{1}{1-2t}$$

$$= (1-2t)^{-1}, \text{ for } t < 1/2$$

(ii) Mean of X

we know that mean is first derivative of the Moment generating function with respect to t at $t = 0$

$$M'_X(t) = \frac{d}{dt}\left[(1-2t)^{-1}\right]$$

$$= -1 \cdot (1-2t)^{-2}(-2)$$

$$= 2(1-2t)^{-2}$$

at $t = 0$

$$E(X) = M'_X(0) = 2(1-2(0))^{-2} = 2(1)^{-2}$$

$$= 2$$

(iii) variance of X

$$\text{var}(X) = E(X^2) - [E(X)]^2$$

for $E(X^2) = M''_X t = \frac{d}{dt}\left(2(1-2t)^{-2}\right)$

$$= 2(-2) \cdot (1-2t)^{-3}(-2)$$

$$= 8(1-2t)^{-3}$$

$$M''_X(0) = 8(1-2(0))^{-3}$$

$$= 8(1)^{-3}$$

$$= 8$$

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Now $\text{Var}(X) = E(X^2) - [E(X)]^2$

$= 8 - (2)^2$

$= 8 - 4$

$= 4$

1. Determine the binomial distribution for which the mean is 4 and variance is 3.

Given that Mean $\mu = np = 4$ — ①

Variance $\sigma^2 = npq = 3$ — ②

from $\frac{②}{①}$ we get $\frac{npq}{np} = \frac{3}{4}$

$\Rightarrow q = \frac{3}{4}$

As we know that in Binomial distribution $p+q=1$

$p = 1-q = 1-\frac{3}{4} = 1/4$

Now from ① we get $n\left(\frac{1}{4}\right) = 4$

$n = 16$

Now By Binomial distribution

$P(X=x) = {}^{n}C_{r}\, p^{n-r} q^{r}$

$= {}^{16}C_{r}\left(\frac{1}{4}\right)^{16-r}\left(\frac{3}{4}\right)^{r}$

2. Out of 800 families with 5 children each how many would you expect to have (a) 3 boys (b) 5 girls (c) either 2 or 3 boys (d) at least one boy (Assume equal probabilities of boys and girls).

Probability of having a boy, $p = \frac{1}{2}$

hence $q = 1-p = 1/2$

no. of children = 5

no. of families = 800

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(i) P (Having 3 boys) = P(x = 3)

$P(x=3) = {}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2$

$= 10 \times \frac{1}{25} = 0.3125$

no. of families that are expected to have 3 boys = $800 \times 0.3125 = 250$

(ii) P (Having 5 girls) = P(x = 0)

$P(x=0) = {}^5C_0 \left(\frac{1}{2}\right)^5$

$= \frac{1}{32}$

no. of families that are expected to have 5 girls = $800 \times \frac{1}{32} = 25$

(iii) P (Having 2 or 3 boys) = P(x = 2) + P(x = 3)

$= {}^5C_2 \left(\frac{1}{2}\right)^5 + {}^5C_3 \left(\frac{1}{2}\right)^5 = 0.625$

∴ no. of families having either 2 or 3 boys = $800 \times 0.625 = 500$

(iv) P (Having atleast one boy) = $P(x \geq 1) = 1 - P(x = 0)$

$= 1 - {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5$

$= 1 - 0.03125$

$= 0.96875$

No. of families that are expected to have atleast one boy

$= 800 \times 0.96875 = 775.$

3. The average number of homes sold by the Realty company is 2 homes per day. Determine the probability that will be sold tomorrow (i) exactly 3 homes (ii) in between 0 and 2, both inclusive.

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Given that

average rate of succes per period of day $= 2 = \lambda$

Now P(x=

from poisson distribution

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

(i) Exactly 3 homes

$$P(x=3) = \frac{e^{-2} \cdot 2^3}{3!} = \frac{0.1353 \times 8}{6} = 0.1805$$

(ii) In between 0 and 2, both inclusive

$$P(0 \le x \le 2) = P(x=0) + P(x=1) + P(x=2)$$

$$= \frac{e^{-2} \cdot 2^0}{0!} + \frac{e^{-2} \cdot 2^1}{1!} + \frac{e^{-2} \cdot 2^2}{2!}$$

$$= e^{-2}\left[1 + 2 + 2\right]$$

$$= 5e^{-2} = 0.67667$$

4. The average number of phone calls per minute coming into a Switch board between 2 p.m and 4 p.m is 2.5. determine the probability that during one particular minute there will be

(i) 4 or fewer (ii) More than 6 calls.

Given the average number of phone calls $\lambda = 2.5$

From poisson distribution $P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$

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(i) P (4 or fewer calls) $= P(x \leq 4)$

$P(x \leq 4) = P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4)$

$= e^{-2.5}\left[\frac{(2.5)^0}{0!} + \frac{(2.5)^1}{1!} + \frac{(2.5)^2}{2!} + \frac{(2.5)^3}{3!} + \frac{(2.5)^4}{4!}\right]$

$= 0.082\left[1 + 2.5 + \frac{6.25}{2} + \frac{15.625}{6} + \frac{39.0625}{24}\right]$

$= 0.082 \times 10.857$

$= 0.08902$

(ii) P (More than 6 calls) $= P(x > 6) = 1 - P(x \leq 6)$

$= 1 - e^{-2.5}[P(x=0) + P(x=1) + \cdots + P(x=6)]$

$= 1 - 0.082\left[10.857 + \frac{(2.5)^5}{5!} + \frac{(2.5)^6}{6!}\right]$

$= 1 - 0.082[10.857 + 0.8138 + 0.3391]$

$= 1 - (0.082 \times 12.009)$

$= 0.052$

5. If the probability that an individual suffers a bad reaction due to certain injection is 0.001. determine the probability that out of 2000 individuals (i) exactly 3 and (ii) More than 2 individuals will suffer a bad reaction.

Given that number of individuals $(n) = 2000$

probability of bad reaction $p = 0.001$

Now mean $\lambda = np = 2000 \times 0.001 = 2$

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From poisson distribution

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

(i) Exactly 3 individuals

$$P(X=3) = \frac{e^{-2} \cdot 2^3}{3!} = \frac{0.13534 \times 8}{6} = \frac{1.08272}{6} = 0.1805$$

(ii) More than 2 individuals

$$P(x>2) = 1 - P(x \leq 2)$$

$$= 1 - [P(x=0) + P(x=1) + P(x=2)]$$

$$= 1 - e^{-2}\left[\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!}\right]$$

$$= 1 - 0.67667$$

$$= 0.32333$$

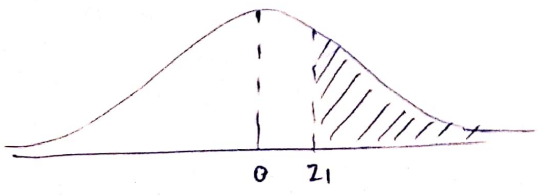
6. If the masses of 300 students are normally distributed with mean 68 kgs and standard deviation 3 kgs, how many students have masses (i) greater than 72 kg (ii) less than or equal to 64 kg.

Given that Mean $\mu = 68$

Standard deviation $\sigma = 3$

(i) $P(x \geq 72) = P(z > 72)$

When $x_1 = 72$; $z_1 = \frac{72-68}{3} = 1.33$



Area $= 0.5 - A(z_1)$

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$= 0.5 - A(1.33)$

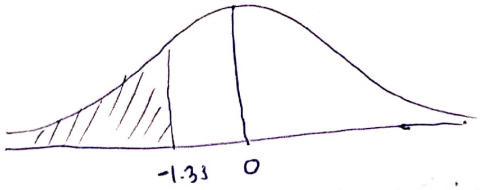
$= 0.5 - 0.4082$

$= 0.0918$

no. of students with more than 72 kgs $= 300 \times 0.0918 = 27.54 \approx 28$

(ii) $P(x \leq 64)$

when $x_1 = 64$ $\quad z_1 = \frac{64-68}{3} = \frac{-4}{3} = -1.33$



Area $= 0.5 - A(z_1)$

Area $= 0.5 - A(z_1)$

$= 0.5 - A(1.33)$ $\quad [\because A(-z_1) = A(z_1)$

$= 0.5 - 0.4082$

$= 0.0918$

$\therefore$ no. of students less than or equal to 64 kgs are $300 \times 0.0918 = 27.54 \approx 28$

7. A company pays its employees an average wage of ₹350 per hour with a standard deviation of ₹30. If the wages are approximately normally distributed, compute the propotion of the workers getting (i) wages between ₹325 and ₹360 per hour, (ii) wages is more than ₹335, (iii) at least wage is ₹380

Given that Mean $(\mu) = 350$

Standard deviation $(\sigma) = 30$

We know $z = \frac{x-\mu}{\sigma}$

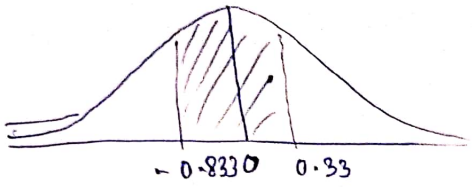
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(i) getting wages between ₹325 and ₹360

$P(325 \leq x \leq 360)$

when $x_1 = 325$; $z_1 = \frac{325-350}{30} = \frac{-25}{30} = -0.833$

when $x_2 = 360$; $z_2 = \frac{360-350}{30} = \frac{10}{30} = 0.33$



Area = $A(z_1) + A(z_2)$

Now $P(-0.83 \leq z \leq 0.33) = A(-0.833) + A(0.33)$

$= A(0.833) + A(0.33)$

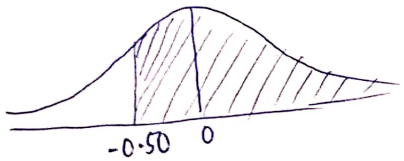
$= 0.2967 + 0.1256$

$= 0.4223$

(ii) getting wages more than ₹335

$P(x > 335)$

when $x = 335$ $z = \frac{335-350}{30} = -0.50$



$A = 0.5 + A(z)$

Area $= 0.5 + A(0.50)$

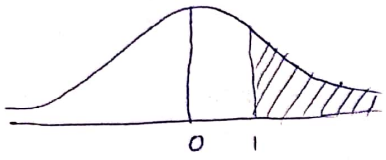
$= 0.5 + 0.1916$

$= 0.6916$

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(iii) getting at least a wage of ₹ 380

when $x = 380$; $z = \frac{380-350}{30} = \frac{30}{30} = 1.$



Area $= 0.5 - A(z)$

Area $= 0.5 - A(1)$

$= 0.5 - 0.3413$

$= 0.1587$

8. If X is poisson variate such that $3P(x=4) = \frac{1}{2} P(x=2) + P(x=0)$

find (i) Mean of X (ii) $P(x \leq 2)$

From the poisson distribution

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

Given $3P(x=4) = \frac{1}{2} P(x=2) + P(x=0)$

$$3\left(\frac{e^{-\lambda} \cdot \lambda^4}{4 \times 3 \times 2}\right) = \frac{1}{2}\left(\frac{e^{-\lambda} \cdot \lambda^2}{2!}\right) + \left(\frac{e^{-\lambda} \cdot \lambda^0}{0!}\right)$$

$$\frac{\lambda^4}{8} = \frac{\lambda^2}{4} + 1$$

$$\lambda^4 = 2\lambda^2 + 8$$

$$\lambda^4 - 2\lambda^2 - 8 = 0$$

$$(\lambda^2 - 4)(\lambda^2 + 2) = 0$$

$\lambda^2 = 4$ (or) $\lambda^2 = -2$ (no real)

$\lambda^2 = 4$

$\lambda = 2$

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(i) Mean $np = \lambda = 2$

(ii) $P(x \leq 2) = P(x=0) + P(x=1) + P(x=2)$

$$= \frac{e^{-2} \cdot 2^0}{0!} + \frac{e^{-2} \cdot 2^1}{1!} + \frac{e^{-2} 2^2}{2!}$$

$$= e^{-2} \left[\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!} \right]$$

$$= e^{-2} (1 + 2 + 2)$$

$$= 5e^{-2}$$

$$= 5 (0.135335)$$

$$= 0.676675$$

9. If X is a normal variate with mean 30 and standard deviation 5. Find (i) $P(26 \leq x \leq 40)$, (ii) $P(x \geq 45)$.

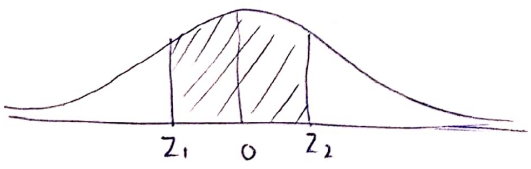
Given that mean $\mu = 30$

standard deviation $\sigma = 5$

(i) To find $P(26 \leq x \leq 40)$

when $x_1 = 26$; $z_1 = \frac{26-30}{5} = \frac{-4}{5} = -0.80$

when $x_2 = 40$; $z_2 = \frac{40-30}{5} = \frac{10}{5} = 2$



$P(26 \leq x \leq 40) = P(z_1 \leq z \leq z_2) = A(z_1) + A(z_2)$

$$= A(0.8) + A(2)$$

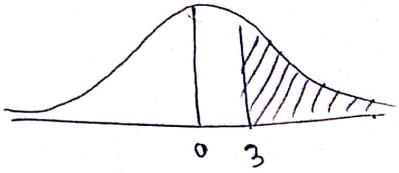
$$= 0.2881 + 0.4772$$

$$= 0.7653$$

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(ii) To find $p(x \geq 45)$

when $x = 45$; $z = \frac{45-30}{5} = \frac{15}{5} = 3$



Area $= 0.5 - A(2)$

$P(x \geq 45) = P(z > z_1) = 0.5 - A(3)$

$= 0.5 - 0.4987$

$= 0.0013$

10. The length of time a person speaks over phone follows exponential distribution with mean 6. what is the probability that the person will talk for (i) More than 8 minutes (ii) between 4 and 8 minutes

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